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Motor Sizing Worksheet: Peak, RMS and Margin Checks

For OEM designers and specifying engineers selecting PMDC, geared or BLDC motors for new industrial machines

Complete each step before contacting suppliers. Record every assumption, because the shortlist must be re-checked with the chosen motor's own rotor inertia, torque constant and resistance before the specification is released.

Step 1 — Define the duty

- [ ] List every motion segment: distance or angle, time, and required speed.
- [ ] Load masses and forces for the worst-case product, not the typical one.
- [ ] On-time, rest time and cycles per hour for the worst-case shift.
- [ ] Agreed load-growth allowance (10–25 % is a reasonable start), applied to the load.
- [ ] Maximum ambient at the motor, altitude, enclosure and IP requirement.
- [ ] Nominal and minimum supply voltage at the motor terminals.

Step 2 — Map the transmission and its efficiency

- [ ] Ratio of every stage, screw lead, drum or pulley diameter.
- [ ] Spur or helical gearing: about 95–98 % per stage.
- [ ] Worm gearing: about 40–90 % depending on ratio and lead angle.
- [ ] Ball screw: around 90 %. Trapezoidal lead screw: about 30–50 %.
- [ ] Use lower efficiencies at breakaway for screws and worm gears.
- [ ] Lead screw torque = force × mean diameter / 2 × tan(lead angle + friction angle); self-locking if lead angle is below friction angle.

Step 3 — Torque at the motor shaft for every segment

- [ ] Segment torque = (load torque + load inertia × load acceleration) / (ratio × efficiency) + rotor inertia × motor acceleration.
- [ ] Start segment: use breakaway torque together with acceleration.
- [ ] Include deceleration, powered lowering and any holding requirement.
- [ ] Use the rotor inertia of the motor actually being considered; repeat if the frame changes.
- [ ] Record the motor speed required in each segment.

Step 4 — Peak, RMS and duty type

- [ ] Peak torque = largest segment torque.
- [ ] RMS torque = square root of [sum of (torque² × time) / total cycle time], rests included.
- [ ] Add start transients if starts are frequent.
- [ ] RMS method valid only when each on-period is short compared with the thermal time constant.
- [ ] Fan-cooled motors cool less when stopped: count only part of the rest time (half is a cautious allowance).
- [ ] Choose duty type: S1 continuous, S2 short-time or S3 intermittent periodic, from the real cycle.

Step 5 — Environment, supply and margins

- [ ] Ambient factor, copper-loss estimate (upper bound): square root of [(rated winding temp − ambient) / (rated winding temp − 40 °C)]. With about 30 % fixed losses, typical of small motors, use about 0.95 at 45 °C and 0.91 at 50 °C.
- [ ] Thermal: RMS torque at 75–90 % of the continuous rating after ambient correction.
- [ ] Peak: no more than about 80 % of the torque available at the controller current limit.
- [ ] Speed: loaded speed = (minimum voltage − current × Ra − brush drop) / Ke, with 5–10 % headroom.
- [ ] Gearbox: output torque within rating after a service factor of 1.0–2.0 for shock and hours.
- [ ] Peak current within the motor's permitted peak (magnet demagnetisation limit for PMDC).

Step 6 — Verify, and avoid oversizing

- [ ] Re-run Steps 3–5 with the shortlisted motor's datasheet values.
- [ ] If RMS sits far below the rating, check whether a smaller frame meets every margin.
- [ ] Check stall torque × ratio against gearbox and machine strength; set the current limit accordingly.
- [ ] Confirm brake, encoder, mounting and connector requirements with the supplier.
- [ ] Document all assumptions in the motor specification.

Worked example from the article: lead-screw lift table, 150 kg payload, 20 mm/s, 9:1 gearbox, Tr20×4 screw

Sizing step	Motor-shaft figure	Comment
Naive: 150 kg × g × 20 mm/s	0.10 Nm (29 W)	Suggests a 40 W motor
+ platform and 20 % growth (210 kg)	0.15 Nm	Still looks light
+ screw (40 %) and gearbox (94 %)	0.385 Nm	Running raise torque
+ breakaway (screw 31 %)	0.50 Nm	At start
+ rotor acceleration (0.5 s ramp)	0.59 Nm	Peak requirement
RMS over 60 s cycle, starts included	0.20 Nm	Thermal requirement
÷ 0.95 for 45 °C, ÷ 0.8 thermal margin	0.26 Nm	Minimum continuous rating
Peak ÷ 0.8	0.74 Nm	Minimum current-limited torque
Speed at 21.6 V and 0.385 Nm	2,940 rpm	9 % above the 2,700 rpm needed

Efficiency ranges and margins are typical engineering values; confirm them against supplier data for your components. Globe Scott engineers can review your sizing through www.globescott.net/contact-us/.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.